



General Assembly

Distr.: General
29 August 2025

Original: English

Human Rights Council

Sixtieth session

8 September–3 October 2025

Agenda items 2 and 3

Annual report of the United Nations

High Commissioner for Human Rights and
reports of the Office of the High Commissioner
and the Secretary-General

Promotion and protection of all human rights, civil,
political, economic, social and cultural rights,
including the right to development

The right to privacy in the digital age

Report of the Office of the United Nations High Commissioner for Human Rights*

Summary

The present report, submitted pursuant to Human Rights Council resolution 54/21, contains a discussion of challenges and risks with regard to discrimination and the unequal enjoyment of the right to privacy associated with the collection and processing of data. It provides an outline of the applicable normative framework, insights into the discriminatory and unequal impacts of data-driven technologies in key sectors and an analysis of contributing factors and resulting challenges. It also contains recommendations on how to prevent digitalization from perpetuating or deepening discrimination and exclusion.

* The present report was submitted to the conference services for processing after the deadline so as to include the most recent information.



I. Introduction

1. The present report is submitted pursuant to Human Rights Council resolution 54/21 in which the Council requested the Office of the United Nations High Commissioner for Human Rights (OHCHR) to prepare a report on challenges and risks with regard to discrimination and the unequal enjoyment of the right to privacy associated with the collection and processing of data to identify and clarify related human rights principles, safeguards and best practices, and to present the report to the Council at its fifty-seventh session. Due to the current liquidity crisis affecting the Secretariat, the presentation of the report was postponed to the sixtieth session.
2. The digital transformation of societies is a defining development of current times, changing economies, political environments and culture. It relies on data collection and processing at an unprecedented scale. The Internet, in particular, appears as a limitless source of data that can be collected, shared, analysed, used and transformed into the main building blocks of applications across almost all domains of human life. Advances in artificial intelligence were built on those processes and further intensified the search for more data. In parallel, over the past two decades, it has become increasingly apparent that the benefits and harms of digitalization are unevenly distributed. Discrimination and inequality stemming from data collection and processing are widespread and can be observed in all sectors of society: image generators that produce pictures only of light-skinned doctors, for example, even when expressly prompted to be more inclusive. Women not receiving job advertisements for management positions. A person being labelled a terrorist based on discriminatory algorithmic profiling. Black people wrongfully arrested because of inaccurate facial recognition. A parent with a disability wrongly investigated for child abuse because of embedded bias in the risk modelling algorithmic systems of child protection agencies. Women facing prosecution for exercising their sexual and reproductive health rights, based on data collected by period tracker applications. Farm workers not receiving welfare payments because their fingerprints cannot be read by biometric identification systems due to decades of manual labour.
3. The present report provides insights into such phenomena, their impacts and contributing factors and offers suggested pathways for overcoming and preventing such discrimination and inequality. It builds on previous reports published by the Secretary-General and OHCHR that addressed many of the complex issues of digitalization and datafication¹ and reflects responses received to the call for inputs issued by OHCHR.²

II. Normative framework

A. International human rights law

4. The right to privacy as a human right is recognized in article 12 of the Universal Declaration of Human Rights, in article 17 of the International Covenant on Civil and Political Rights and in other international and regional human rights instruments.³ The right to privacy plays a pivotal role in the balance of power between the State and the individual and is foundational to the exercise of rights in a democratic society.⁴ Its importance for the

¹ [A/HRC/27/37](#), [A/HRC/39/29](#), [A/HRC/43/29](#), [A/HRC/48/31](#), [A/HRC/51/17](#), [A/HRC/59/32](#) and [A/79/296](#).

² See <https://www.ohchr.org/en/calls-for-input/2025/right-privacy-digital-age>.

³ See article 16 of the Convention on the Rights of the Child, article 14 of the International Convention on the Protection of the Rights of All Migrant Workers and Members of Their Families, article 22 of the Convention on the Rights of Persons with Disabilities, article 10 of the African Charter on the Rights and Welfare of the Child, article 11 of the American Convention on Human Rights and article 8 of the Convention for the Protection of Human Rights and Fundamental Freedoms (European Convention on Human Rights).

⁴ [A/HRC/39/29](#), para. 11.

enjoyment and exercise of other human rights both online and offline⁵ is growing in an increasingly data-centric world.

5. The right to privacy is vital for human dignity and necessary for the protection of human autonomy and personal identity.⁶ It protects a sphere of autonomous development, interaction and liberty, with or without interaction with others, free from State intervention and from excessive unsolicited intervention by other actors.⁷

6. The dimensions of privacy important in the context of data collection and processing include: informational privacy, covering information that exists or can be derived about a person and their life and decisions based on that information;⁸ decisional privacy, which ensures an individual's ability to make autonomous life choices; bodily privacy, which protects a person's control over their own body and protects from unwanted physical contact, observation or intrusion; and communicational and associational privacy (the privacy of communicating and associating with others).

7. Any State interference with the right to privacy must be compatible with international human rights law and therefore meet the standards of legitimate purpose, legality, necessity and proportionality.⁹

8. The right to privacy can enable the enjoyment of other rights.¹⁰ Thus, discrimination and the unequal enjoyment of the right to privacy in the context of data collection and processing may affect other rights, including the rights to life, liberty and security, freedom of expression,¹¹ peaceful assembly,¹² freedom of movement, effective remedy, health, education and housing.

9. The right to equality and the prohibition against discrimination are core principles of international human rights law, affirmed in article 7 of the Universal Declaration of Human Rights and in articles 2 and 26 of the International Covenant on Civil and Political Rights and in other treaties. Non-discrimination is an intrinsic principle of international law, existing not only inherently in the obligation of States to ensure respect, protection and fulfilment of human rights without discrimination but also, as enshrined in article 26 of the Covenant, as a guarantee of equality before the law and equal protection of the law.

10. Differential treatment is discriminatory if it has the purpose or effect of nullifying or impairing the recognition, enjoyment or exercise by all persons, on an equal footing, of rights, unless the justification for differentiation is reasonable and objective.¹³ This will include an assessment as to whether the aim and effects of the measures or omissions are legitimate, aimed at promoting the general welfare in a democratic society, and proportionate.¹⁴

11. Human rights treaties prohibit discrimination on the grounds of race, colour, sex, language, religion, political or other opinion, national or social origin, property, birth or other status. The reference to "other status" in treaties indicates that that list is not exhaustive and that grounds of comparable nature can be incorporated. Additional grounds are commonly recognized when they reflect the experience of social groups that are vulnerable and have

⁵ Committee on the Rights of the Child, general comment No. 25 (2021) on children's rights in relation to the digital environment, paras. 67 and 68; and [A/HRC/39/29](#), para. 11.

⁶ Committee on the Rights of the Child, general comment No. 25 (2021), para. 67; and European Court of Human Rights, *Goodwin v. United Kingdom*, Application No. 28957/95, Judgment, 11 July 2002, para. 90.

⁷ See [A/HRC/39/29](#) and *Coeriel and Aurik v. Netherlands* (CCPR/C/52/D/453/1991).

⁸ [A/HRC/39/29](#), para. 5.

⁹ [A/HRC/48/31](#), para. 8. See also General Assembly resolution 77/211 and Human Rights Council resolution 54/21.

¹⁰ See Human Rights Council resolution 54/21 and [A/HRC/48/31](#).

¹¹ See submission from Center for Studies on Freedom of Expression and Access to Information.

¹² See submission from Pete Fussey and Daragh Murray.

¹³ Human Rights Committee, general comment No. 18 (1989) on non-discrimination in economic, social and cultural rights, paras. 6 and 13; and Committee on Economic, Social and Cultural Rights, general comment No. 20 (2009) on non-discrimination in economic, social and cultural rights, para. 13.

¹⁴ Committee on Economic, Social and Cultural Rights, general comment No. 20 (2009), para. 13.

suffered and continue to suffer marginalization.¹⁵ The Committee on Economic, Social and Cultural Rights, for example, has recognized age, nationality, marital and family status, sexual orientation and gender identity, disability, health status and economic and social situation as protected grounds.¹⁶ Both direct and indirect forms of differential treatment can constitute prohibited discrimination.

12. Article 20 (2) of the International Covenant on Civil and Political Rights requires that any advocacy of national, racial or religious hatred that constitutes incitement to discrimination, hostility or violence be prohibited by law. Articles 19 and 20 of the Covenant complement each other. Article 19 (3) establishes that any restriction to the right to freedom of expression must be provided by law and be necessary for respect of the rights or reputation of others, or for the protection of national security or of public order, or of public health or morals.¹⁷

13. Business enterprises have a responsibility to respect human rights, meaning that they should avoid infringing on the human rights of others and address adverse human rights impacts with which they are involved, in line with the Guiding Principles on Business and Human Rights.¹⁸ The Guiding Principles require businesses to identify, prevent and mitigate adverse rights impacts stemming from or being linked to their business operations, and reflect the obligations of States to protect against human rights abuses within their jurisdiction by business enterprises and to ensure access to effective remedy when abuses occur.

14. In a growing number of resolutions, the General Assembly and the Human Rights Council have recognized that the discriminatory impacts of the datafication of societies pose a serious challenge to human rights.¹⁹ Discrimination risks in the artificial intelligence context are recognized in the Global Digital Compact, which contains a commitment to develop data and metadata standards designed to prevent and address bias, discrimination or human rights violations and abuses throughout the data life cycle (see para. 42 below).

B. Other relevant normative frameworks

15. In response to the ever-increasing capabilities of data-driven technologies, numerous legal frameworks for data collection and processing have been developed. They include the African Union Convention on Cyber Security and Personal Data Protection and the Convention for the Protection of Individuals with regard to Automatic Processing of Personal Data of the Council of Europe. Moreover, national data protection laws have been adopted in many States, even though considerable gaps remain.

16. Discrimination and inequality are addressed in those and other data-related frameworks in several ways, including by granting certain categories of data – the processing of which carries increased risks of discrimination – a heightened degree of protection.²⁰ Other instruments directly prohibit discrimination via data processing.²¹

17. Recent efforts have been focused on building legal frameworks for artificial intelligence. Equality and non-discrimination are central concerns addressed in such frameworks, such as the Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and Rule of Law and laws at the regional and national levels, such as the Artificial Intelligence Act of the European Union and the Basic Act on the

¹⁵ Ibid., para. 27.

¹⁶ Ibid., paras. 27–35.

¹⁷ Human Rights Committee, general comment No. 34 (2011) on the freedoms of opinion and expression, para. 50.

¹⁸ See Human Rights Council resolution 17/4.

¹⁹ See General Assembly resolutions 78/187, 78/213 and 78/265 and Human Rights Council resolution 54/21.

²⁰ See African Union Convention on Cyber Security and Personal Data Protection and the modernized version of the Convention for the Protection of Individuals with regard to Automatic Processing of Personal Data of the Council of Europe (Amending Protocol, Council of Europe Treaty Series No. 223).

²¹ See General Data Protection Act of Brazil, Guidelines for the regulation of computerized personal data files and Principles on Personal Data Protection and Privacy. See also [ST/SGB/2024/3](#).

Development of Artificial Intelligence and Establishment of Trust of the Republic of Korea.²² The United Nations Educational, Cultural and Scientific Organization has adopted a recommendation on the ethics of artificial intelligence, which underscores the centrality of non-discrimination in the responsible use of artificial intelligence.

III. Data-driven discrimination and inequality

A. Disparate impacts in key domains

18. Section III contains a description of major issues concerning data-driven discrimination and inequality as they unfold in key sectors. It should be noted that similar issues emerge in sectors beyond those addressed in the present report, as well as in contexts of armed conflict; a discussion of those was precluded by space limitations.

19. Problems often arise when automated decision-making systems, predictive analytics and biometric recognition systems embed biases or are deployed in contexts that make it likely that they will single out or disproportionately affect certain groups. Biometric recognition systems, for example, including facial recognition systems, have proven accuracy issues, depending on skin tone and gender. Even if such outcomes can be mitigated by developers,²³ other problems remain. Surveillance technologies have historically been deployed in ways that disproportionately affect low-income and marginalized communities and are sometimes used to target certain groups.²⁴ Biometric recognition systems provide the information needed to profile individuals on the basis of their ethnicity, race, gender and other personal characteristics.²⁵ Biometric recognition systems have also been used to identify and track peaceful protestors and dissidents, amounting to discrimination on the basis of political opinion. The deployment of such powerful surveillance technology may also chill the exercise of public freedoms and other rights.²⁶ Previous reports of the High Commissioner contain further analysis of such issues.²⁷

1. Law enforcement and criminal justice

20. Increasingly, data-driven predictive tools are being used both by the police and courts.²⁸ They include tools used to predict where future crimes may occur and systems to determine the risk of recidivism or to flag individuals as risks or likely suspects. Such systems depend on large quantities of data, including historical and live data, criminal records, arrest records, crime statistics, records of police interventions in specific places, social media posts and metadata, communications data and location data. It is well documented that such systems' outputs often display significant biases, often as a consequence of historical biases in law enforcement and criminal justice systems, adversely affecting racial, ethnic and religious minorities in particular.²⁹ That is in addition to concerns arising from the use of biometric recognition, as referenced above.

²² See <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>; and https://cset.georgetown.edu/wp-content/uploads/t0625_south_korea_ai_law_EN.pdf. Many other States are discussing artificial intelligence-related bills; see submission from Colombia.

²³ See https://pages.nist.gov/frvt/reports/demographics/nistir_8429.pdf.

²⁴ See <https://www.aclu.org/documents/aclu-comment-facial-recognition-and-biometric-technologies-eo-14074-13e>; and submissions from Access Now and Human Rights Myanmar.

²⁵ See [A/HRC/44/57](#), paras. 39 and 40; and [A/HRC/48/31](#).

²⁶ See [A/HRC/44/24](#), [A/HRC/48/31](#) and [A/HRC/51/17](#); and submission from Pete Fussey and Daragh Murray. See also [A/HRC/48/76](#) and [A/75/590](#).

²⁷ [A/HRC/44/24](#), [A/HRC/48/31](#) and [A/HRC/51/17](#).

²⁸ See [A/79/296](#).

²⁹ See Committee on the Elimination of Racial Discrimination, general recommendation No. 36 (2020) on preventing and combating racial profiling by law enforcement officials. See also the conference room paper of the Human Rights Council on the promotion and protection of the human rights and fundamental freedoms of Africans and of people of African descent against excessive use of force and other human rights violations by law enforcement officers, para. 93, available on the web page of the Council, at <https://www.ohchr.org/en/hr-bodies/hrc/regular-sessions/session47/list-reports>.

2. Migration

21. People on the move are often subject to particularly pervasive and intrusive data collection and processing practices, with serious implications for human rights. Infrastructure to manage migration is relying increasingly on extensive data-driven technologies, including artificial intelligence, automated algorithmic decision-making systems, predictive analytics and related digital technologies.³⁰ Such technologies can be used for many purposes, including detection at physical borders, the determination of risks as defined by the State in algorithmic risk assessments,³¹ the verification of identities and the management of migrants' presence in a country. Data collected, stored, processed and shared can include biometric data, health data, social media data, educational attainments, employment status, nationality, location data, information about personal and professional networks, family status, criminal records, financial data and data about political and other activities. Data are collected from the people concerned, State agencies, private companies, including data brokers, international organizations and other States. Governments have been creating increasingly integrated, interoperable systems and databases, permitting access to data for a vast array of agencies for a broad range of purposes.³² Based on such systems and databases, the authorities determine who can enter a country or must leave and who can have access to benefits and services. While those objectives are legitimate, adverse effects often disproportionately hit people with certain national or ethnic backgrounds because, for example, of bias embedded in data-driven decision-making systems and, at times, discriminatory profiling practices.³³

3. Social benefits

22. Data-driven technologies are becoming an increasingly central element of how States manage and deliver essential services, including welfare and public health. The objectives for such systems include increased efficiency, the accurate delivery of services and the detection of fraud. They are used to determine programme eligibility and benefit amounts, enforce income and asset limits and verify identity. They are also deployed to flag situations in which interventions are necessary, for example where abusive situations may occur. There is great potential to leverage algorithmic decision-making and support to advance such important goals. The Special Rapporteur on extreme poverty and human rights has noted, for example, that the automation of benefits could be a key way to address the non-take-up of social benefits, as it could reduce administrative complexity for potential recipients.³⁴

23. There are significant risks for the equal enjoyment of the right to privacy and other rights. Access to social services has often been linked to deteriorating privacy guarantees for people on low incomes, with applicants and beneficiaries expected to provide extensive information about their private lives, including being subjected to intrusive measures. That phenomenon disproportionately affects women and members of racial or ethnic minorities and can be perpetuated by digital technologies.³⁵ A former Special Rapporteur on extreme poverty and human rights referred to the use of unfettered data-matching to expose, survey and punish welfare beneficiaries as a potential "digital welfare dystopia".³⁶

24. The data used are submitted in part by applicants, gleaned in part from other sources, including various government databases and social media, or purchased from commercial providers.³⁷ The use of algorithmic decision-making or support systems can result in the

³⁰ See [A/HRC/48/76](#) and [A/75/590](#).

³¹ Including even pseudoscientific emotion recognition systems that fail to understand cultural and other nuances and may even operationalize eugenicist theories of phrenology. See [A/HRC/48/31](#) and submission from Access Now.

³² See <https://www.ohchr.org/sites/default/files/2023-09/Digital-Border-Governance-A-Human-Rights-Based-Approach.pdf>; and submission from AsyLex.

³³ See <https://www.ohchr.org/sites/default/files/2023-09/Digital-Border-Governance-A-Human-Rights-Based-Approach.pdf>.

³⁴ See [A/HRC/50/38](#).

³⁵ See Virginia Eubanks, *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor* (New York, St. Martin's Press, 2017); and Khiara M. Bridges, *The Poverty of Privacy Rights* (Redwood City, California, Stanford Law Books, 2017).

³⁶ See [A/74/493](#).

³⁷ [A/HRC/48/31](#), para. 30.

cutting off of a main source of income, the withholding of access to medical treatment and the denial of basic housing; they can even trigger criminal investigations and prosecutions. The denial of benefits can have far-reaching negative effects on individuals' lives, such as insolvency, the loss of housing and deteriorating health. As a consequence, the people most in need of support may not benefit from automation but rather be excluded.³⁸ Given the risks of such negative impacts, it is essential that such systems be especially reliable and closely monitored.

25. There are many reports, however, of algorithmic systems deployed in the context of social benefits that were poorly designed, inaccurate or plainly wrong and used in deeply discriminatory ways. For example, the Auditor General of the State of Michigan, United States of America, reviewed 22,000 fraud determinations from an automated detection system deployed by the state's unemployment insurance scheme and found that 93 per cent did not actually involve fraud.³⁹ Automated identity verification systems that put benefits applications on hold, sometimes for months or years, have caused thousands of vulnerable people to forgo or receive reduced payments, while burdening staff with additional verification work.⁴⁰ Proxy means testing, often used to enable automated assessments when there is no access to individualized data, has drawn criticism for failing to capture the nuanced realities of poverty and vulnerability. In several settings, high levels of exclusion and inclusion errors have been documented, with exclusion errors reaching, in some programmes, values as high as 70 and 93 per cent.⁴¹

4. Health

26. The digitalization of healthcare services has great potential to directly improve people's lives. By reducing the need for physical assessment, virtual care offers opportunities to reach remote communities and other people who face barriers in physical access to healthcare facilities. The use of artificial intelligence and machine learning enables the analysis of large data sets for prediction, forecasting, drug and vaccine development and medical diagnostics.⁴²

27. At the same time, digital health applications touch upon some of the most intimate and sensitive aspects of people's lives, implicating core privacy dimensions. As the Special Rapporteur on the right of everyone to the enjoyment of the highest attainable standard of physical and mental health has underscored, digital health technologies can perpetuate racism, sexism, ableism or discrimination based on sexual orientation or gender identity, among others.⁴³ For example, medical tools and applications can perform differently depending on factors, such as skin tone, ethnic background, age and gender. A review commissioned by the Government of the United Kingdom of Great Britain and Northern Ireland found evidence that artificial intelligence-enabled devices used to diagnose skin cancer could perform more poorly for patients with darker skin tones and raised concerns about the underdiagnosis of heart disease in women and of glaucoma in people of Asian descent.⁴⁴ Artificial intelligence-driven medical case management can also display stark biases. One such product, used on more than 200 million people in hospitals to predict which patients would be likely to need extra medical care, identified light skin-toned patients far more often than Black patients with a comparable health status.⁴⁵

³⁸ See [A/HRC/50/38](#) and Sebastian Smart, "Algorithmic discrimination in Latin American welfare States", Carr Center Discussion Paper, No. 2024-05 (Cambridge, Massachusetts, Harvard Kennedy School, 2024).

³⁹ United States Court of Appeals, *Cahoo et al. v. SAS Analytics Inc. et al.*, No. 18-1296 (6th Cir. 2019).

⁴⁰ Kevin De Liban, *Inescapable AI: The Ways AI Decides How Low-Income People Work, Live, Learn, and Survive* (Los Angeles, TechTonic Justice, 2024), p. 59.

⁴¹ See Smart, "Algorithmic discrimination".

⁴² See [A/HRC/53/65](#).

⁴³ *Ibid.*

⁴⁴ See <https://www.gov.uk/government/publications/equity-in-medical-devices-independent-review-final-report>. See also submission from Privacy International.

⁴⁵ See Ziad Obermeyer and others, "Dissecting racial bias in an algorithm used to manage the health of populations", *Science*, vol. 366, No. 6464 (October 2019).

28. Multiple studies have indicated that health data, including those collected via mobile health applications, are often shared without the consent or even the knowledge of the people concerned. That is particularly concerning when information includes gender identity, sexuality, location or health status, including when such information can be inferred from other data. Women seeking reproductive healthcare can be exposed to particular risks.⁴⁶ For example, a broad range of health applications collect vast amounts of data that may be accessible to law enforcement agencies in jurisdictions in which abortion is criminalized.⁴⁷ Other applications, such as dating applications, have been vectors for investigating and criminalizing LGBTQI+ persons.⁴⁸ Health insurance companies obtaining access to sensitive health or genetic data, for example by purchasing data from data brokers, can lead them to deny coverage or essential care based on health status or even simply potential health status.⁴⁹

5. Digital public infrastructure

29. Efforts by States to deploy foundational digital systems, including identity and payment systems, that enable the delivery of essential services in such domains as welfare, health, education and finance at scale have gained remarkable traction in recent years. Such systems have great potential to facilitate increased inclusion but can also constitute an obstacle to the enjoyment of human rights or even contribute to rights violations and abuses.⁵⁰ As a key layer in allowing data-intensive transactions across domains, digital public infrastructure can enable arbitrary surveillance. Because of the scale and nature of the data amassed, any data breaches can affect people in profound ways. There is also strong evidence for discriminatory and exclusionary effects associated with digital public infrastructure, in particular digitized identification systems.⁵¹

30. Such effects may materialize in various ways. For example, there can be considerable obstacles to enrolling in identification systems, including costly or difficult registration processes, a failure to reach remote areas and displaced and stateless people, customs and practices disadvantaging minority groups or women and girls and fear among vulnerable sections of the population that the system may be used against them. System design and technical failures can also contribute to disparate outcomes, such as non-inclusive design choices (e.g. data categories that fail to correspond to self-identification) and a lack of stable Internet connectivity. Factors in using digital public infrastructure can also play a major role in creating unequal outcomes, including excessive identification requirements, function creep (i.e. the expansion of use requirements across initially unforeseen sectors) and a lack of access to or recognition of alternative means of identification.

6. Surveillance

31. As highlighted in previous reports, an array of surveillance activities carried out by both States and businesses constitute major threats to the right to privacy and other human rights. Most public and many private activities can be monitored, quantified and analysed. Surveillance cameras are ubiquitous in many places and increasingly equipped with biometric recognition technology. So-called smart cities are being set up in all regions of the world, creating highly complex systems of constant data collection and processing, enabled by increasingly advanced sensors. Online spaces are under constant watch by authorities and companies that collect massive amounts of information about Internet users, profile individuals and analyse networks. Such data can be used for both legitimate and illegitimate purposes, including in conflict settings. There has been growing attention to how data collected and processed, including through surveillance, are used in armed conflict, and the

⁴⁶ See submissions from AI & Equality and Association for Progressive Communications.

⁴⁷ See submission from Privacy International.

⁴⁸ See Gwen Aviles, “Tinder introduces safety feature to protect LGBTQ travelers”, NBC News, 24 July 2019; and submission from ILGA Asia.

⁴⁹ See Casey Ross and Bob Herman, “Denied by AI: how Medicare Advantage plans use algorithms to cut off care for seniors in need”, *STAT*, 13 March 2023; and Marshall Allen, “Health insurers are vacuuming up details about you – and it could raise your rates”, *ProPublica*, 17 July 2018.

⁵⁰ See Universal DPI Safeguards Framework for guidance on a range of safeguards against rights violations and other risks, available at <https://www.dpi-safeguards.org>.

⁵¹ See [A/HRC/43/29](#) and submission from Access Now.

severe impacts on civilians and civilian objects.⁵² Ubiquitous surveillance can be abused for racial profiling and to identify and track people for discriminatory purposes, for example because of their political beliefs. Even when serving legitimate objectives, from managing complex urban environments to facilitating access to public services, such surveillance often fails to meet the proportionality requirement. It poses a serious challenge to the right to privacy and the freedoms at the heart of open, democratic societies.⁵³

7. Online information

32. The online information environment is shaped by data-driven processes, which can contribute to discriminatory outcomes. An important study of search engines exposed widespread racist outputs,⁵⁴ a concerning result which is also reflected in other forms of discriminatory and hateful expression. The targeting of content in social media builds on vast amounts of data collected about every user, which can include discriminatory targeting. It has been reported that advertisers can, on some platforms, select their target audience according to criteria that, at least indirectly, correspond to prohibited grounds, for example by excluding members of certain racial or ethnic groups from seeing housing or job advertisements or by deliberately targeting vulnerable people with exploitative products such as gambling and payday loans.⁵⁵ Engagement-focused curation algorithms, targeting content based on data about previous browsing patterns, can also lead to the amplification of hate speech and xenophobic content. At the same time, studies have found that automated content moderation, including hate speech detection systems, can be biased and thereby disproportionately limit content from members of minorities.⁵⁶ Such systems tend to be especially unreliable, and potentially harmful, for non-dominant languages.⁵⁷ Generative artificial intelligence image generators have also been shown to produce highly prejudiced outputs.⁵⁸

B. Common challenges

1. Issues relating to systems and their design

(a) Technical design and decision-making

33. Disparate treatment can have its origins in the design process of a data-driven system, starting with early decisions around what it is supposed to achieve and how that will be measured.⁵⁹ If the goal, for example, is to optimize the deployment of police by predicting high occurrences of crime, the type of crime included will influence who will be affected; property crime may more often be committed by poor people while tax fraud is a crime associated with wealth.

34. The quality of data used to develop and operate data-driven technologies may be another major problem. Often, the data sets used are tainted by inaccuracies and discriminatory labelling. For example, an investigation into a data set widely used for training machine vision systems revealed countless instances of arbitrary labelling, incoherences and

⁵² See Russell Buchan and Asaf Lubin, eds., “The rights to privacy and data protection in times of armed conflict”, *International Review of the Red Cross*, vol. 105, No. 923 (June 2023); and Robin Geiss and Henning Lahmann, “Protection of data in armed conflict”, Working Paper, Geneva Academy of International Humanitarian Law and Human Rights, February 2021.

⁵³ See [A/HRC/51/17](#).

⁵⁴ See Safiya Umoja Noble, *Algorithms of Oppression* (New York, NYU Press, 2018).

⁵⁵ See <https://verfassungsblog.de/targeted-ad>; and submission from AI & Equality.

⁵⁶ See <https://aclanthology.org/P19-1163.pdf>.

⁵⁷ See <https://ecnl.org/publications/algorithmic-gatekeepers-impacts-llm-content-moderation-civic-space-and-human-rights>.

⁵⁸ See Ananya, “AI image generators often give racist and sexist results: can they be fixed?”, *Nature*, 19 March 2024.

⁵⁹ See Solon Barocas and Andrew D. Selbst, “Big data’s disparate impact”, *California Law Review*, vol. 104, No. 3 (June 2016); and European Union Agency for Law Enforcement Cooperation (Europol), *AI Bias in Law Enforcement: A Practical Guide* (Luxembourg, Publications Office of the European Union, 2025), pp. 22–50.

plainly racist, misogynistic and otherwise hateful and discriminatory labels; eventually, the database operator removed 600,000 images.⁶⁰ Incompleteness and non-representativeness of data are major vectors of disparate outputs of algorithmic systems.⁶¹ For example, historically disadvantaged groups are often under- or overrepresented in data sets, skewing any analysis of the sample. Consider a predictive policing system that uses historical arrest data that may reflect past racially biased policing practices, posing a risk of steering future predictions in the same, biased direction. That, in turn, may lead to future overpolicing, causing more arrests within the same group and thereby creating a dangerous feedback loop.⁶² The common assumption of the representativeness of data is rarely justified and perhaps more often incorrect than correct.⁶³ Moreover, data that can be adequate in certain contexts may not be fit for purpose for others. For example, if a model was trained on crime records from one State where a certain type of crime had high prevalence, it would fail in another where such crime may be less common.⁶⁴

35. Another related issue is the effect of proxies on the outputs of algorithmic systems. Even if care is taken that a data set does not include direct data about protected characteristics, other data points can stand in as indicators for such characteristics.⁶⁵ For example, postcodes or certain given names can correlate with membership in a group and thus become proxies that skew data analysis to the disadvantage of the people concerned.

(b) Transparency and accountability

36. Data-driven tools present unprecedented challenges with respect to the transparency of decision-making and accountability for those decisions. Two challenges predominate. First, government and private actors often fail to provide adequate information about their data-driven products, services and tools. Individuals affected by such systems and the general public often cannot know when data-driven systems make decisions affecting them, let alone how. In such circumstances it is challenging, if not impossible, to meaningfully scrutinize data-driven decision-making. Second, the data environment, algorithms and models are immensely complex. The output of machine-learning systems, including large language models, is often difficult or impossible to explain, a phenomenon frequently referred to as the “black box” problem.⁶⁶ Consequently, accountability in cases where such systems cause harm remains difficult, for example when a person harmed must prove fault and causality for her damages claims. That, in turn, limits incentives for developers, operators and users of data-driven automated systems to act lawfully and in non-discriminatory ways.

(c) Scalability

37. The scalability of data-driven tools can dramatically increase the negative effects of even small error rates and bias.⁶⁷ The problems ingrained in such systems can reach hundreds, thousands or even millions of people. That is a key difference from individual biases.

(d) Inferences and predictions

38. The ability of data-driven applications to make inferences and predictions brings significant risks. Seemingly innocuous data, often not even related to a specific person, are widely used to predict human behaviour, draw conclusions about people’s inner world, their

⁶⁰ See <https://excavating.ai>; and <https://image-net.org/update-sep-17-2019>.

⁶¹ See Barocas and Selbst, “Big data’s disparate impact”.

⁶² Committee on the Elimination of Racial Discrimination, general recommendation No. 36 (2020), para. 33.

⁶³ See Barocas and Selbst, “Big data’s disparate impact”.

⁶⁴ Arvind Narayanan and Sayash Kapoor, *AI Snake Oil: What Artificial Intelligence Can Do, What It Can’t, and How to Tell the Difference* (Princeton, New Jersey, Princeton University Press, 2024), p. 52.

⁶⁵ See Barocas and Selbst, “Big data’s disparate impact”; and Europol, *AI Bias in Law Enforcement*, pp. 24 and 32.

⁶⁶ See Ariel Bleicher, “Demystifying the black box that is AI”, *Scientific American*, 9 August 2017. Nevertheless, such systems do not have to be entirely inscrutable; see <https://doi.org/10.1145/3351095.3372873>.

⁶⁷ A/HRC/48/31, para. 18.

interests, opinions and character traits, and determine sensitive personal characteristics, such as health status and sexual orientation. Such inferences and predictions can result in decisions affecting people's rights, sometimes in a fully automated way.⁶⁸ For example, browser histories can reveal information about someone's sexual life, health status or political views, which in turn could be used to deny health insurance coverage or make someone the object of political persecution. Adequate safeguards against such practices are urgently needed, including by updating relevant legislation, such as data protection, anti-discrimination, consumer protection and sectoral laws.⁶⁹

2. Human, structural and institutional bias

39. Data-driven risks and harms regularly affect marginalized people and those in vulnerable situations. Differentiated adverse impacts on members of certain groups often intersect; discrimination and exclusion can be fully grasped and addressed only through an intersectional approach.⁷⁰ For example, discrimination based on socioeconomic status often intersects with racial or ethnic discrimination. Women journalists face online harassment campaigns disproportionately more often than their male counterparts, and in very specific gender-focused ways.

40. Systemic bias often reflects deep-seated human bias and assumptions, including those carried over longer periods. Institutional racism and sexism are common examples. Such bias can be perpetuated, for instance, in the decision to deploy certain digital tools in contexts in which it is far more likely to affect historically disadvantaged populations. It is well documented, for example, that intrusive surveillance technologies and other data-intensive tools disproportionately target minorities or migrants.⁷¹ A vast body of research points to stereotypes (e.g. unemployed people do not want to work, foreigners are criminals) and ideological positions (e.g. poverty is an individual moral failure) being a common undercurrent of such decisions.⁷²

41. While some such biases may unconsciously seep into various decisions in the design, deployment and operation of data processing systems, sometimes discrimination, in breach of international human rights law, can be the very goal of such systems. For example, systems could directly discriminate by identifying dissidents through automatic scans of social media posts or by targeting particular religious communities to expose them to systemic surveillance.

42. At times, the perceived objectivity of data and data-driven tools hides deeply unfair practices in both providing public services and offering products and services on the market. New, untested or scientifically unfounded technologies are at times rolled out first with regard to people who lack the means and capacity to challenge such deployments, such as migrants or low-income people.⁷³ Data-driven technologies have also been used as powerful tools for oppression by enabling the identification, surveillance and censoring of human rights defenders, protesters and dissidents.

3. Role of the private sector

43. The dominant role of the private sector in driving technological development and roll-out presents particular challenges. Business enterprises have a responsibility to respect

⁶⁸ See [A/HRC/48/31](#).

⁶⁹ Ibid. and submission from Privacy International.

⁷⁰ See General Assembly resolution 78/187; Committee on the Elimination of Discrimination against Women, general recommendations No. 35 (2017) on gender-based violence against women, updating general recommendation No. 19, and No. 39 (2022) on the rights of Indigenous women and girls; Committee on the Elimination of Racial Discrimination, general recommendation No. 36 (2020); and United Nations network on racial discrimination and protection of minorities, *Guidance Note on Intersectionality, Racial Discrimination & Protection of Minorities* (Geneva, OHCHR, 2022). See also submissions from Association for Progressive Communications and Media Monitoring Africa.

⁷¹ See OHCHR and University of Essex, "Digital border governance: a human rights-based approach", September 2023; and [A/75/590](#).

⁷² See Eubanks, *Automating Inequality*; Bridges, *The Poverty of Privacy Rights*; and Noble, *Algorithms of Oppression*.

⁷³ [A/75/590](#), paras. 38–42.

human rights across their operations, while States have a duty to protect against human rights abuses by business enterprises.⁷⁴ The Guiding Principles on Business and Human Rights call upon States to apply a smart mix of the regulatory and policy measures available to them to protect against human rights harms related to the products and services of technology companies, including regulatory measures and accompanying guidance, incentives and transparency requirements.

44. Comprehensively applying that framework poses myriad challenges in the context of digital technologies. Not only are corporate data collection and processing practices often excessive and linked to discrimination, but States face many obstacles to providing effective protections. A lack of capacity to fully assess the offerings of the private sector, for example, may disadvantage some State actors. Pressure to create a supportive and competitive business environment may constrain States' willingness to build the guardrails needed to address human rights concerns.⁷⁵

45. Such challenges may be particularly acute when there is a close State-business nexus in the development and operation of data-driven tools, such as when States rely on private suppliers or enter into public-private partnerships. In such contexts, there is a risk of blurring responsibilities between State and private actors, especially when supply chains are long and complex. Where States financially support, contract or procure from technology companies, they must actively take steps to ensure that the companies that they work with respect human rights. States should use their position to insist on rights-based business conduct and the delivery of products that do not lead to unlawful discrimination. They should oversee the development and deployment of procured data-driven systems and demand and assess information about the data sets and the accuracy and risks of the application. Where risks cannot be mitigated effectively, States should not use, deploy or authorize the products concerned.⁷⁶ Applications that cannot be operated in compliance with international human rights law should be banned.⁷⁷

4. Digital divides

46. Data-driven discrimination and inequality often reflect persistent patterns of exclusion from participation in the digital sphere. In 2024, almost one third of the world's population had no access to the Internet and Internet penetration reached only 27 per cent in low-income countries. Gender divides remain and are particularly pronounced in low-income and lower-middle-income countries. There is also a marked access divide between rural and urban areas. Poverty generally means lower access to the Internet.⁷⁸ There are also strong disparities in digital literacy and skills depending on location, socioeconomic status, gender and other factors. Job markets in digital sectors still are marked by gender disparities. For example, women make up only roughly 30 per cent of the artificial intelligence workforce.⁷⁹

47. Digital applications appear often to be developed with the goal of targeting regions and audiences promising high profits. Certain digital solutions may not be available to people living in low- or middle-income countries, persons with disabilities, minorities or Indigenous Peoples, among others or are unaffordable or of poor quality for the context.⁸⁰

48. The dominance of certain languages in many digital products and services also prevents individuals, often those mentioned in the previous paragraph, from having access to them. Social media user interfaces and policies are available in a limited number of languages. A lack of language support increases the risk of inappropriate data use, as it affects users'

⁷⁴ See <https://www.ohchr.org/sites/default/files/Documents/Issues/Business/B-Tech/b-tech-foundational-paper-state-duty-to-protect.pdf>.

⁷⁵ See [A/HRC/59/32](#) and [A/HRC/59/53](#).

⁷⁶ See [A/HRC/48/31](#).

⁷⁷ Ibid. See also General Assembly resolution 78/213.

⁷⁸ See https://www.itu.int/hub/publication/D-IND-ICT_MDD-2024-4; and <https://pulse.internetsociety.org/en/blog/exploring-the-global-state-of-internet-poverty-in-2024>.

⁷⁹ See Committee on the Elimination of Discrimination against Women, general comment No. 40 (2024) on the equal and inclusive representation of women in decision-making systems; and <https://www.randstad.com/randstad-ai-equity>.

⁸⁰ See submissions from Guatemala and Montenegro.

ability to understand and adjust privacy and security settings.⁸¹ The performance of dominant generative artificial intelligence offerings is consistently high in very few languages. Those limitations affect billions of users and can limit their participation in political discourse, culture and the economic sphere. When certain languages are not resourced online, a wealth of information is not taken into account. That may affect how future products and services will be shaped and may also contribute to inadequate policy responses. It should be noted that strong community efforts have led to the development of websites, tools, resources and a range of language models for low-resource languages;⁸² future work on expanding the digital reach of languages should enhance those efforts.

49. In addition, many States lack sufficient regulatory frameworks, including data protection laws and, even when they are in place, the authorities may have a limited ability to enforce them effectively, especially in low-revenue markets.⁸³ That leaves individuals in those regions particularly vulnerable to human rights violations and abuses, including discriminatory data practices. While many States have enhanced their regulatory frameworks, policies and practices in recent years, further efforts are needed to ensure that the right to privacy is upheld equally across all regions.⁸⁴

IV. Conclusions and recommendations

50. The increasing reliance on data-driven technologies, including artificial intelligence, has brought about significant and growing transformations in societies, changing economies, political environments and cultures. Those developments have introduced new and complex challenges to the enjoyment of the rights to privacy and to non-discrimination and equality. The collection and processing of data, particularly when used to inform automated decision-making, can perpetuate and exacerbate existing inequalities and discriminatory practices – and are already doing so at scale. Those most likely to be victims of those processes are often already marginalized people and those in vulnerable situations.

51. Technological innovation must be anchored in international human rights law and standards. Discrimination and inequality are not marginal issues; they are systemic failures that compromise the integrity of societies, democratic, participatory governance and the realization of human rights for all. Considering the speed with which data-driven technologies are being deployed and the scale of harms that they can bring, those issues must be addressed with the greatest urgency, from the local to the global level. Against that background, signs of a growing backlash against setting guardrails for the development of data-driven technologies under the banner of fostering innovation are deeply worrying. Strong human rights safeguards and guidance are needed to enable innovation that works for everyone.

52. Data-driven technologies reflect the structural inequalities that permeate social, economic and political realities. Addressing those issues requires a comprehensive and intersectional approach that goes beyond the scope of the present report. Nonetheless, there are specific and actionable steps that States, businesses and other stakeholders should take. Many key recommendations have been laid out in previous reports of the High Commissioner and in resolutions of the Human Rights Council.⁸⁵ The following should be read in conjunction with such recommendations. Moreover, the recommended measures should not be seen in isolation but should be part of sustained, comprehensive whole-of-society efforts.

⁸¹ See Nyalleng Moorosi, “Better data sets won’t solve the problem – we need AI for Africa to be developed in Africa”, *Nature*, vol. 636, No. 8042 (December 2024).

⁸² See <https://tehiku.nz/te-hiku-tech>; and <https://theconversation.com/how-ai-could-help-safeguard-indigenous-languages-255359>.

⁸³ See submission from Association for Progressive Communications.

⁸⁴ See <https://techpoint.africa/news/worldcoin-delete-biometric-data-kenya>. See also submissions from Guatemala and Media Monitoring Africa.

⁸⁵ [A/HRC/44/24](#), [A/HRC/48/31](#), [A/HRC/51/17](#), [A/HRC/53/42](#) and [A/HRC/59/32](#); and Human Rights Council resolutions 50/15, 54/21 and 59/11.

53. **OHCHR recommends that States:**

(a) **Establish regulatory frameworks that comprehensively address discrimination and inequality occurring in the context of data collection and processing. Such frameworks should not be based on siloed approaches and should include dedicated legislation on data protection, privacy, freedom of information, anti-discrimination, artificial intelligence and sectoral regulations. States should ensure that any exceptions established by domestic law do not affect negatively the enjoyment of human rights or result in human rights violations, such as wholesale carveouts for law enforcement, migration management or national security;**

(b) **Protect individuals from human rights abuses linked to corporate data collection and processing through the application of a smart mix of regulatory measures and policy incentives;**

(c) **Ensure effective oversight and enforcement capacities, including by establishing and adequately resourcing independent, impartial data protection authorities, anti-discrimination bodies and sectoral oversight;**

(d) **Only use data-driven tools in full compliance with international human rights law, including the rights to privacy, equality and effective remedy and the prohibition of discrimination and ensure that any limitations to human rights adhere to the principles of legality, necessity and proportionality;**

(e) **Condition the procurement of data-driven technologies, products and services on full compliance with non-discrimination standards;**

(f) **Ensure that digital public infrastructure is designed, rolled out, operated and used in ways that uphold the rights to privacy, non-discrimination and equality by establishing data privacy and security standards that are fully compatible with international law, creating strong legal, operational, procedural and technological safeguards to prevent the abuse of such systems, including for unlawful or otherwise arbitrary surveillance, limiting the type of information contained, who has access to it and how it is used and ensuring inclusivity and full accessibility for all;**

(g) **Refrain from making access to public services, including social benefits, healthcare and basic education, conditional upon digital identification without allowing alternative means of identification;**

(h) **Ensure strict compliance with international human rights law, including the principles of necessity and proportionality, when considering the use of biometric technologies, such as for operating digital public infrastructure and in law enforcement contexts;⁸⁶**

(i) **Ensure that victims of human rights violations and abuses, including of the rights to non-discrimination and equality, linked to practices of improper data collection, processing and use have access to effective remedies. With that aim, and given the gravity of the potential harms, consider establishing or expanding legal aid and victim support services as necessary in those contexts;**

(j) **Urgently take all steps necessary to combat digital divides within and between countries, including by accelerating international, regional and domestic efforts to build the infrastructure required for universal meaningful access to the Internet, promoting local content in all relevant languages and enhancing digital literacy;**

(k) **Improve international cooperation, capacity-building, technical assistance and research to enable all countries to develop, deploy and regulate data-driven technologies according to the needs of their populations, grounded in human rights;**

⁸⁶ For details on remote biometric recognition technology, including a recommendation for a moratorium, at least until sufficient safeguards are in place, see [A/HRC/44/24](#), para. 53 (j) (i–v); [A/HRC/48/31](#), para. 59 (d); and [A/HRC/51/17](#), paras. 56 (g) and 57 (d).

(l) Facilitate dialogue and debate about data-driven technologies and their regulation in existing and future intergovernmental and multi-stakeholder processes, including in the context of the Global Digital Compact and the World Summit on the Information Society process, with a view to promoting human rights and sustainable development.

54. OHCHR recommends that States and business enterprises:

(a) Systematically conduct human rights due diligence throughout the life cycle of the data-driven systems that they design, develop, deploy, sell, obtain or operate, with a particular focus on the discriminatory and other disparate impacts of such systems. A key element of their human rights due diligence should be regular, comprehensive human rights impact assessments;

(b) Dramatically increase the transparency of their use of data-driven automated decision-making systems, including by adequately informing the public and affected individuals, enabling independent and external auditing of automated systems and consistently monitoring such systems and their impacts, in collaboration with relevant stakeholders. The more likely and serious the potential or actual adverse human rights impacts, the more transparency is needed;

(c) Ensure that the workforce responsible for the development and deployment of data-driven technologies and the employees using such tools are sensitive to the issues of discrimination and inequality and have the capacity to adequately address such problems. That includes taking decisive steps to ensure workforce diversity and conducting awareness-raising and capacity-building measures, such as training on combating all forms of discrimination, including implicit bias and systemic discrimination, for all those involved in the design, development and deployment of artificial intelligence;

(d) Ensure meaningful engagement with all relevant stakeholders in decisions regarding the development, deployment and use of data-driven systems, in particular affected individuals and groups.

55. OHCHR further recommends that business enterprises:

(a) Make all efforts to meet their responsibility to respect human rights, including through the full operationalization of the Guiding Principles on Business and Human Rights, with a focus on combating discrimination linked to their development, sale or operation of data-driven products and services;

(b) Have a policy commitment in place that sets out their approach to respecting the right to privacy and preventing discriminatory impacts of data collection and processing;

(c) Monitor the effectiveness of actions to mitigate discriminatory impacts and consult with affected groups regularly;

(d) Provide for and participate in effective operational-level grievance mechanisms for digital products and services and provide effective remedies to affected individuals and communities, including in cooperation with State-based judicial mechanisms, where such products or services have caused or contributed to adverse human rights impacts, including discrimination.

56. OHCHR recommends that standard-setting organizations develop and implement standards to combat discrimination in data processing in open processes that include all the steps necessary to ensure the participation of all relevant stakeholders, including members of affected groups, based on the recommendations set out in the report of OHCHR on human rights and technical standard-setting processes for new and emerging technologies.⁸⁷

⁸⁷ [A/HRC/53/42](#).

57. All relevant stakeholders should take proactive and sustained measures to identify, prevent and mitigate bias in data and algorithmic systems, including by:

(a) Investing in research and the development of methodologies to detect, mitigate and eliminate bias in data sets and models;

(b) Improving data collection practices to reduce systemic bias at the source, for example by ensuring that data-collection processes are inclusive and representative of diverse populations and by involving affected communities in decisions about what data are collected and how they are labelled and used;

(c) Institutionalizing bias mitigation in development workflows, for example by systematically integrating impact assessments;

(d) Fostering interdisciplinary collaboration and knowledge-sharing, including by promoting collaboration between technologists, social scientists, ethicists and human rights experts.
